

Feasibility Study for a Waste Management System for the University Campus

Assessment of MRF Viability and Proposal for an Aerobic Composting Pit System

1. Introduction

This feasibility report evaluates the suitability of setting up a waste-processing facility on the university campus. Based on updated waste generation data, it has been determined that a Material Recovery Facility (MRF) is not viable due to insufficient dry waste. However, the campus generates adequate biodegradable waste to justify an **aerobic composting pit system**, which is the recommended solution.

2. Campus Waste Generation Assessment

2.1 Residential Households

- 50 households × 3.6 members ≈ **180 residents**
 - Per household weekly waste:
 - Biodegradable waste: **3.6 kg/week**
 - Dry waste: **0.5 kg/week**
 - Total weekly waste: **205 kg/week**
 - **Daily waste generation ≈ 29 kg/day**
-

2.2 Hostels

- Kitchen biodegradable waste: **47.5 kg/week**
 - Dry waste: **11 kg/week**
 - Total weekly waste: **58.5 kg/week**
 - **Daily waste ≈ 8.4 kg/day**
-

2.3 Academic Areas, Students, and Grounds

- Total enrolled students: **~3,959**

- Daily presence: **~2,000 students**
- Estimated student-generated mixed waste: **0.1 kg/day per student** → ~200 kg/day
- Additional staff waste + leaf litter: **20–50 kg/day**

3. Total Campus Waste Estimate

Source	Daily Waste (kg/day)
Residential	~29
Hostels	~8.4
Students + Campus Areas	200–250
Total Waste	≈ 237–287 kg/day (≈ 0.24–0.29 TPD)

4. Dry Waste Quantities and MRF Feasibility

4.1 Actual Dry Waste Generated

- Residential dry waste: ~3.5 kg/day
- Hostel dry waste: ~1.6 kg/day
- Academic dry waste: limited portion of total mixed waste

Total daily dry waste rarely exceeds 15–25 kg/day.

4.2 Why an MRF Is Not Feasible

A functional MRF requires **0.5–1 TPD** of dry waste to justify equipment, manpower, and operating costs.

Your campus generates **20–40 times less** than the minimum requirement.

Key reasons MRF is not feasible:

- Insufficient dry waste volume
- High operational and labour cost for very low throughput
- Recyclables generated are too small for mechanized sorting
- Not economically sustainable at campus scale

Conclusion:

A Material Recovery Facility (MRF) is NOT viable for the campus.

A small manual dry-waste sorting area is sufficient.

5. Organic Waste Processing Feasibility

5.1 Organic Waste Volume Available

The campus generates **150–240 kg/day of biodegradable waste**, including:

- Kitchen waste
- Canteen leftovers
- Garden leaf litter
- Mess and hostel food scraps

5.2 Why Aerobic Composting Is Ideal

Low cost

Handles mixed organic waste easily

No machinery or electricity required

Produces compost for campus gardens

Scalable with additional pits

Minimal technical expertise required

Therefore, aerobic composting is the **best-suited technology**.

6. Aerobic Composting Pit System Design

6.1 Required Capacity

Daily biodegradable waste: **150–240 kg/day**

Recommended pit volume: **10–15 m³**

6.2 Recommended Pit Layout (3-Pit System)

Pit	Dimensions (L × B × H)	Volume	Purpose
-----	------------------------	--------	---------

Pit 1	2.0 m × 2.0 m × 1.2 m	4.8 m ³	Active filling
-------	-----------------------	--------------------	----------------

Pit 2	2.0 m × 2.0 m × 1.2 m	4.8 m ³	Aerobic curing
-------	-----------------------	--------------------	----------------

Pit 3	2.0 m × 2.0 m × 1.2 m	4.8 m ³	Final maturation
-------	-----------------------	--------------------	------------------

Total volume: **14.4 m³**

Adequate for up to **250 kg/day** organic waste.

6.3 Space Requirement

- **12–20 m²** including working area
 - Roofed shed required to protect from rain
-

7. Composting Process Workflow

Daily Operations

1. Collect segregated organic waste.
2. Remove contaminants (plastic, metal, glass).
3. Add dry leaves/sawdust as carbon source.
4. Spread waste evenly in Pit 1.
5. Turn waste every 3–5 days to maintain aeration.

Processing Timeline

- **Active composting:** 30–40 days
- **Curing stage:** 20–30 days
- **Total cycle:** 45–60 days

Final Output

- Stabilized compost for use in campus gardens and landscaping.
-

8. Infrastructure & Cost Estimate

8.1 Required Infrastructure

- 3–5 RCC composting pits
- Rain-proof shed
- Leachate drainage system
- Sorting table for organic waste
- Tools (shovels, rakes, wheelbarrow)
- PPE for workers
- Signage + instruction boards

8.2 Estimated Costs

Component	Estimated Cost (INR)
Compost pits (3–5 units)	₹60,000 – ₹1,20,000
Shed structure	₹40,000 – ₹80,000
Tools & PPE	₹5,000 – ₹10,000
Leachate management	₹5,000 – ₹15,000
Training & awareness	₹5,000 – ₹10,000
Total Estimated Project Cost: ₹1.2 – ₹2.2 lakh	

9. Implementation Plan

Phase 1: Setup (1–2 Months)

- Allocate land
- Construct pits and shed
- Install leachate drainage
- Arrange tools and signage
- Begin segregation awareness campaign

Phase 2: Pilot Operation (3–6 Months)

- Start operations using Pit 1
- Monitor moisture, odour, and temperature
- Record daily waste quantities
- Adjust carbon (dry leaves) as needed

Phase 3: Full Operation (After 6 Months)

- Establish stable 3-pit rotation
- Produce compost regularly
- Use compost in gardens

Optional Phase 4

- Add a small vermicompost unit using pre-composted material

- Upgrade manual dry waste sorting area
-

10. Risks & Mitigation

Risk	Impact	Mitigation
Poor segregation	Odour, pests	Awareness + clear bin system
Too much moisture	Anaerobic smell	Add dry leaves, turn more frequently
Rainwater entering pits	Leachate overflow	Proper roofing
Lack of trained staff	Poor quality compost	Regular supervision & training

11. Conclusion

The campus produces **sufficient organic waste (150–240 kg/day)** but **insufficient dry waste (15–25 kg/day)** to operate an MRF.

Therefore:

MRF is not feasible.

Aerobic composting pits are fully feasible and recommended.

A **3-5 pit aerobic composting system** provides a simple, low-cost, effective, and sustainable solution for campus waste management.